Experiment - 2

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- **(A)**

  A university administration requires an analytical performance report for each academic unit. Write a PostgreSQL solution to calculate the average marks scored by students grouped by their respective departments. The solution must join the student profile, course subjects, and grade marks tables, returning a breakdown sorted by the highest average marks down to the lowest.

Output

Department	Average Marks
Mathematics	95.00
Computer Science	83.33

Solution:

```
SELECT D.DEPT NAME AS DEPARTMENT,
SCORED), 2) AS AVERAGE MARKS
FROM DEPARTMENTS AS D
JOIN
STUDENTS AS S
ON D.DEPT ID=S.DEPT ID
JOIN
MARKS AS M
ON
S.STUDENT ID = M.STUDENT ID
GROUP BY D.DEPT NAME
ORDER BY AVERAGE MARKS DESC;
```

(B)

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to **merge these datasets** and identify **each unique employee** (by EmpID) along with their **lowest recorded salary** across both systems.

Objective
1. Combine two tables A and B.
2. Return each EmpID with their **lowest salary,** and the corresponding **Ename.**

Table A

EmpID	Ename	Salary
1	AA	1000
2	BB	300

Table B

EmpID	Ename	Salary
2	BB	400
3	CC	100

Output

EmpID	Ename	Salary
1	AA	1000
2	BB	300
3	CC	100

Solution:

```
SELECT EMPID, MIN(ENAME) AS ENAME, MIN(SALARY) AS SALARY
FROM
SELECT * FROM A
UNION ALL
SELECT * FROM B
) AS INTERMEDIATE RES
GROUP BY EMPID;
```

(C)

A multinational organization maintains separate records for employee information and department details. The management wants to identify the employees who receive the highest salary within their respective departments for an annual performance report.
If more than one employee earns the same highest salary in a department, all such employees should be included in the report. The final output should display the **Department Name**, **Employee Name**, and **Salary**, arranged in ascending order of department name.

Note: The employee and department information are maintained in separate normalized tables.

Input Table: **Employee**

ID	NAME	SALARY	DEPT_ID
1	JOE	70000	1
2	JIM	90000	1
3	HENRY	80000	2
4	SAM	60000	2
4	MAX	90000	1

Department

ID	DEPT_NAME
1	IT
2	SALES

OUTPUT

DEPT_NAME	NAME	SALARY
IT	MAX	90000
IT	JIM	90000
SALES	HENRY	80000

Part B: FIND THE 2ND HIGHEST EARNING EMPLOYEE from the same schema.

Solution:

PART A:

```
SELECT D.DEPT NAME, E.NAME, E.SALARY
FROM
EMPLOYEE AS E
JON
DEPARTMENT AS D
ON E.DEPARTMENT ID - D.ID
WHERE E.SALARY IN
SELECT MAX(SALARY)
FROM EMPLOYEE AS EI
WHERE EI .DEPARTMENT ID - E.DEPARTMENT ID
ORDER BY DEPT NAME;
```

PART B:

```
SELECT D.DEPT NAME, E.NAME, E.SALARY
FROM
EMPLOYEE AS E
JON
DEPARTMENT AS D
ON E.DEPARTMENT ID
WHERE E.SALARY
D.ID
SELECT MAX(SALARY)
FROM EMPLOYEE AS EI
WHERE EI .DEPARTMENT ID - E.DEPARTMENT ID
AND EI .SALARY NOT IN
SELECT MAX(SALARY)
FROM EMPLOYEE AS E2
WHERE E2.DEPARTMENT ID - EI .DEPARTMENT ID
ORDER BY DEPT NAME;
```